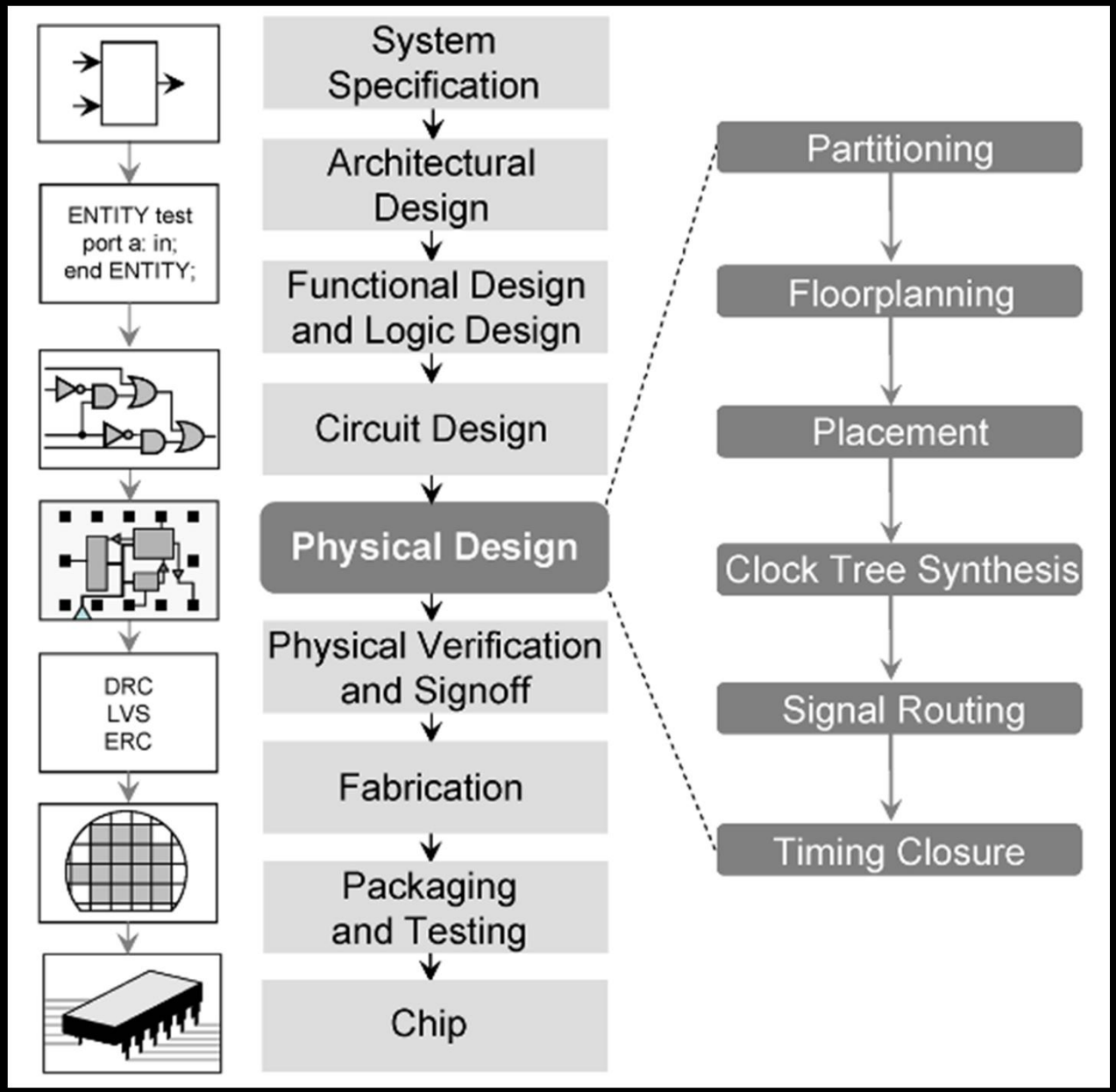
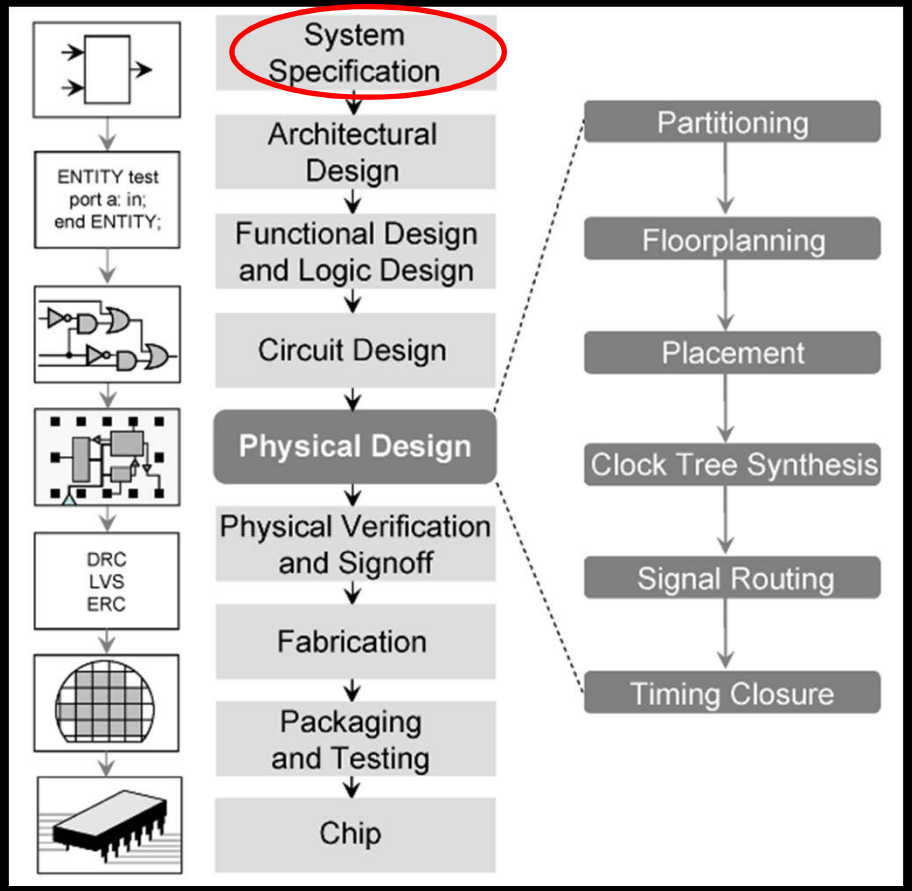
Fluxo de Projeto para Desenvolvimento

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity and_or_top is
    Port ( INA1 : in  STD_LOGIC;    -- AND gate input
           INA2 : in  STD_LOGIC;    -- AND gate input
           OA   : out STD_LOGIC;    -- AND gate output
end and_or_top;

architecture Behavioral of and_or_top is
begin
    OA <= INA1 and INA2;    -- 2 input AND gate
end Behavioral;
```

Fluxo de Projeto para Desenvolvimento

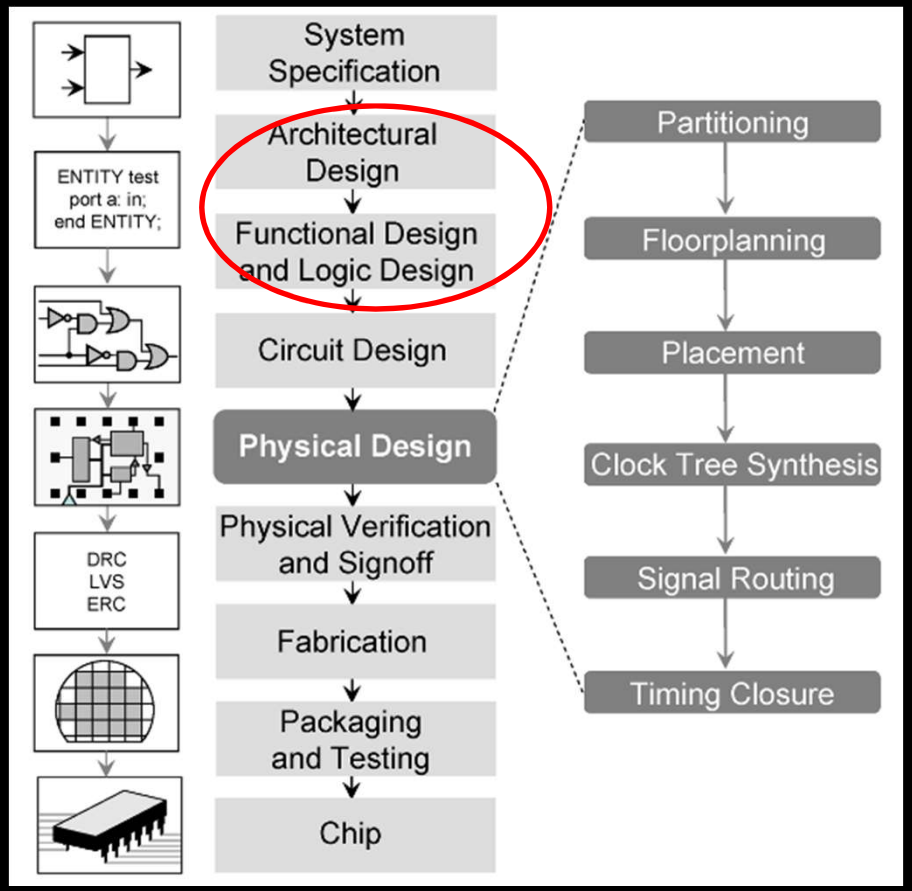


Fluxo de Projeto para Desenvolvimento



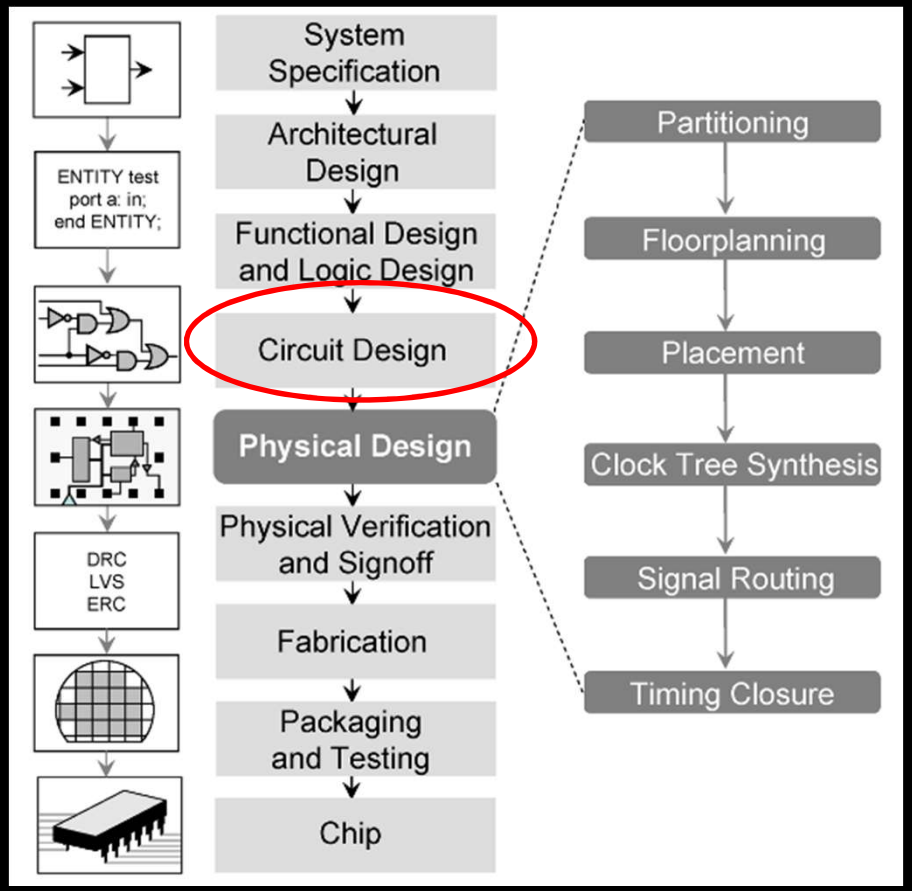
Descrição da funcionalidades e características relacionadas ao uso

Fluxo de Projeto para Desenvolvimento



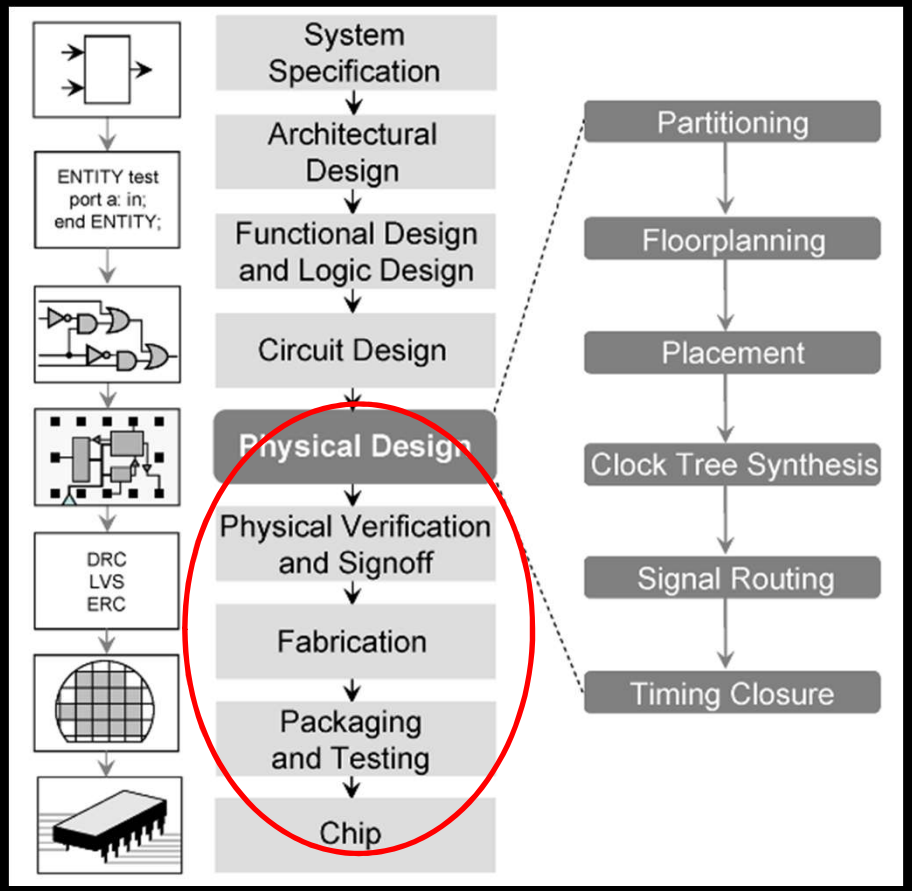
Descrição funcional em Alto nível com HDL

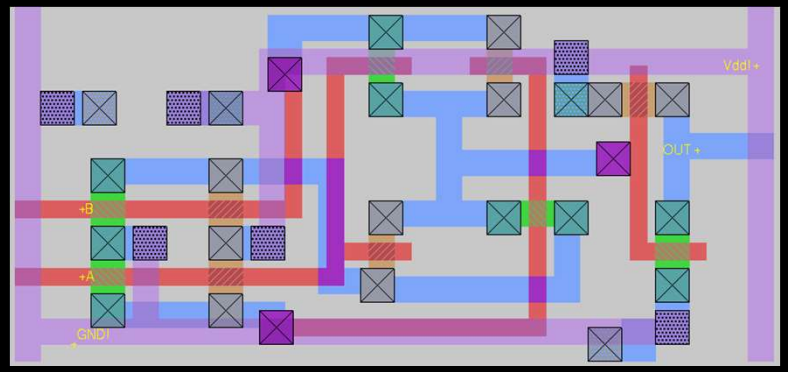
Fluxo de Projeto para Desenvolvimento

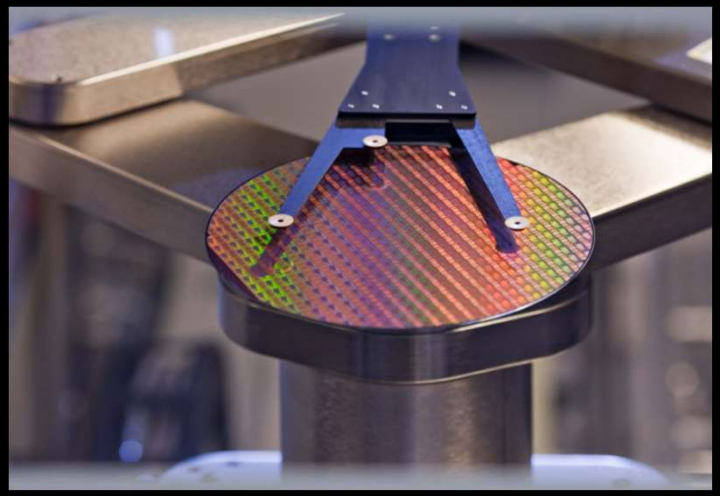


Descrição a nível de portas

Fluxo de Projeto para Desenvolvimento

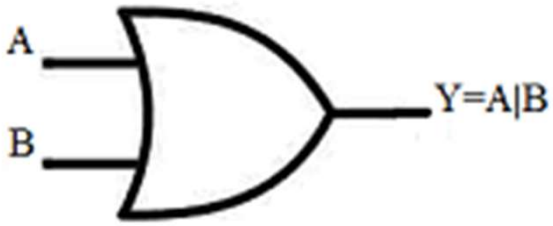






Fluxo de Projeto para Desenvolvimento

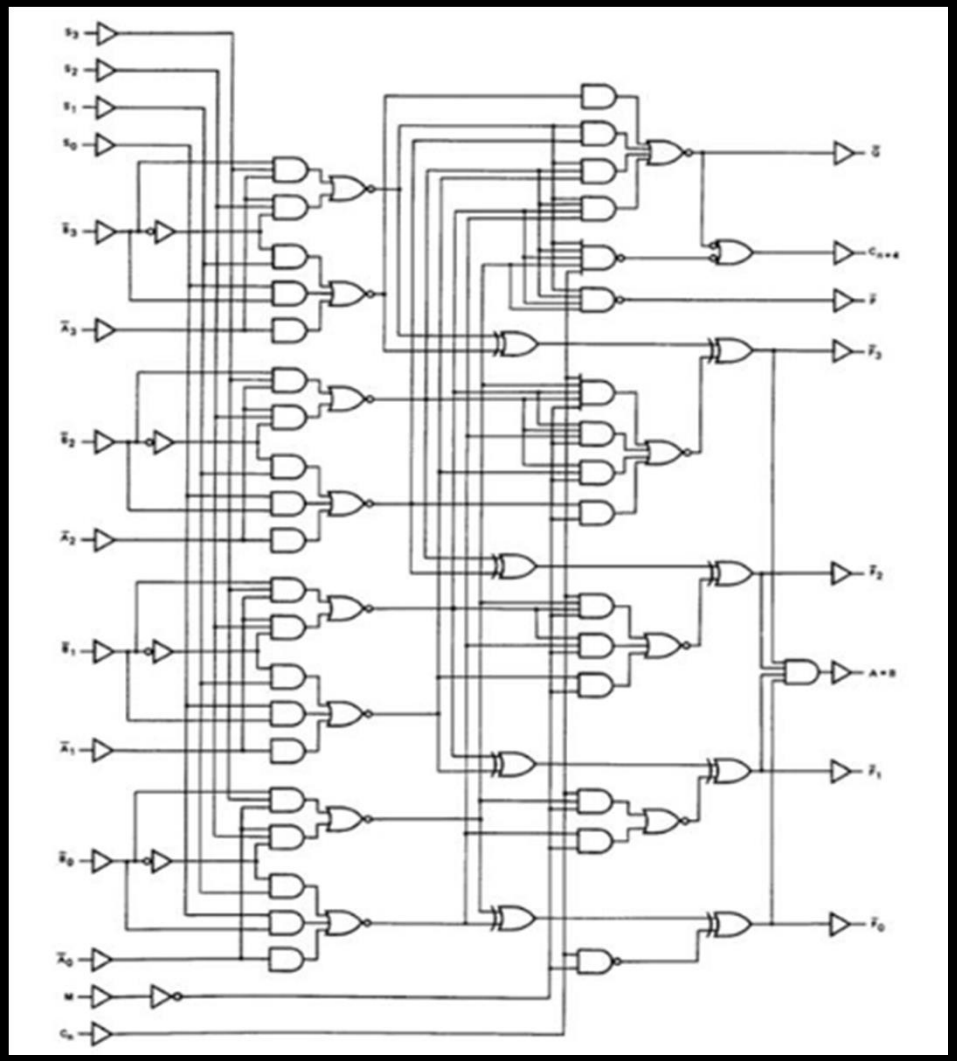
Testes Hardware



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Fluxo de Projeto para Desenvolvimento

Testes Hardware



Fluxo de Projeto para Desenvolvimento

Testes

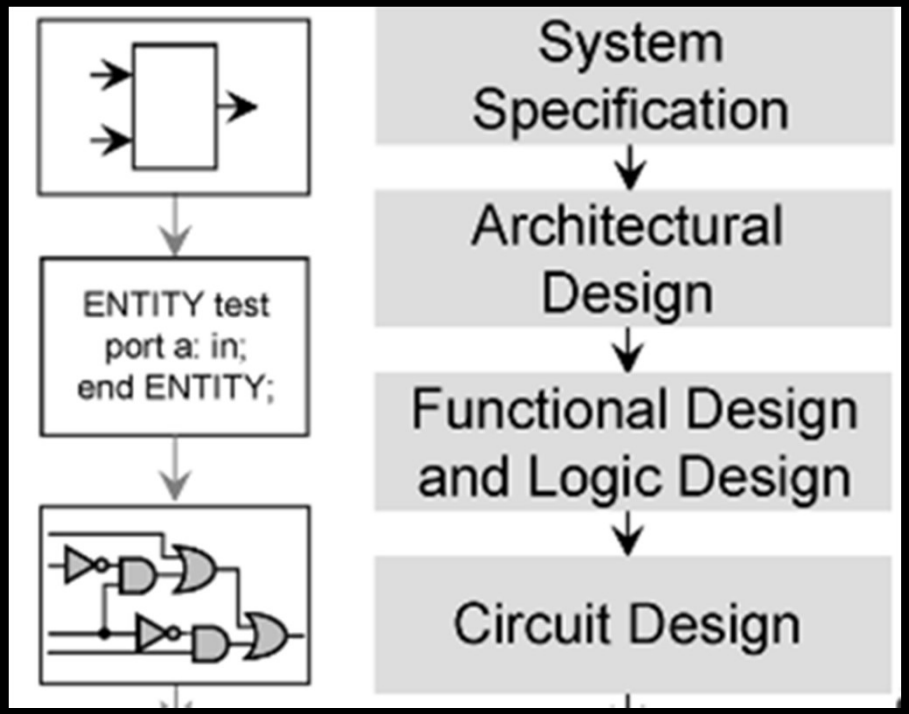
Testbench.vhd

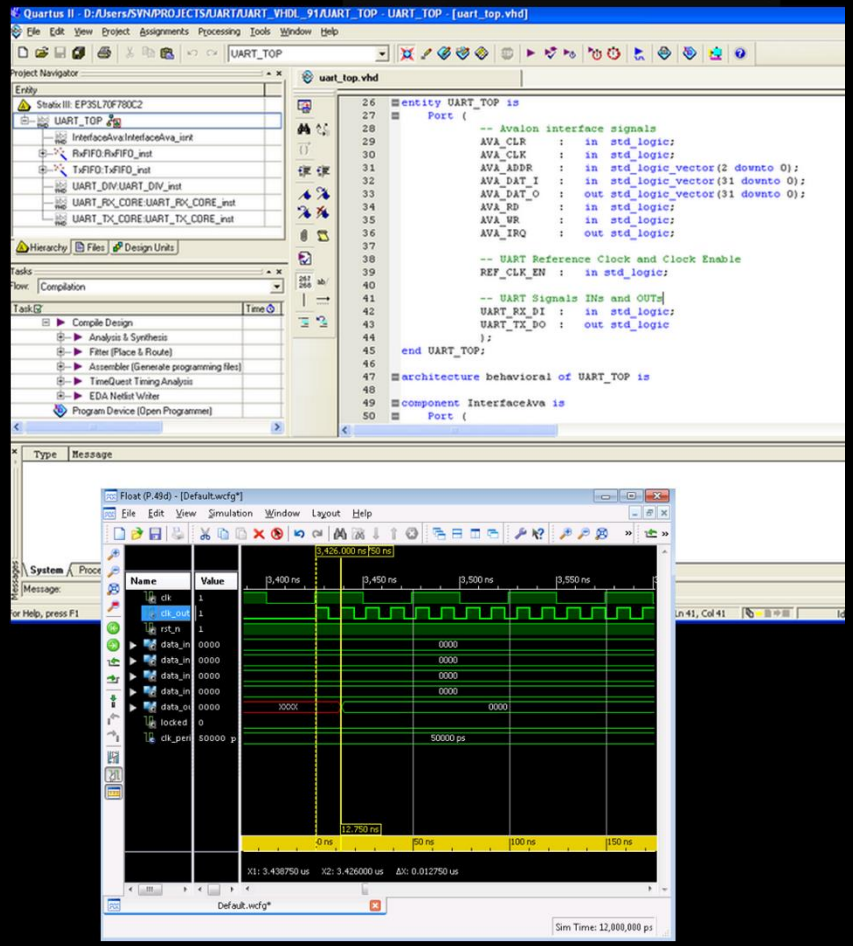
```
process
begin
  a <= '0';
  b <= '0';
  wait for 1 ns;
  assert(q='0') report "Fail 0/0" severity error;

  a <= '0';
  b <= '1';
  wait for 1 ns;
  assert(q='1') report "Fail 0/1" severity error;
```

Or_gate.vhd

```
-- Simple OR gate design
library IEEE;
use IEEE.std_logic_1164.all;

entity or_gate is
port(
  a: in std_logic;
  b: in std_logic;
  q: out std_logic);
end or_gate;

architecture rtl of or_gate is
begin
  process(a, b) is
  begin
    q <= a or b;
  end process;
end rtl;
```

Fluxo de Projeto para Desenvolvimento





Fluxo de Projeto para Desenvolvimento

Software , Fabricantes

XILINX Applications Products

Xilinx - Adaptable. Intelligent. › Developer Tools › ISE Design Suite › ISE WebPACK Design

ISE WebPACK Design Software

ISE Design Suite for Windows 10 - 14.7

Last Updated February 2018

This version of ISE Design Suite only supports Spartan®-6 FPGA
recommends Vivado® Design Suite for new design starts with V
Kintex®-7, Artix®-7, and Zynq®-7000. For other products, ISE De
Suite 14.7 is available by clicking the appropriate link on the left.

14.7 Window 10 (TAR/GZIP - 6.94 GB)

MD5 SUM Value : b639754d9130fa01469cf7e47fa28d84

System Requirements

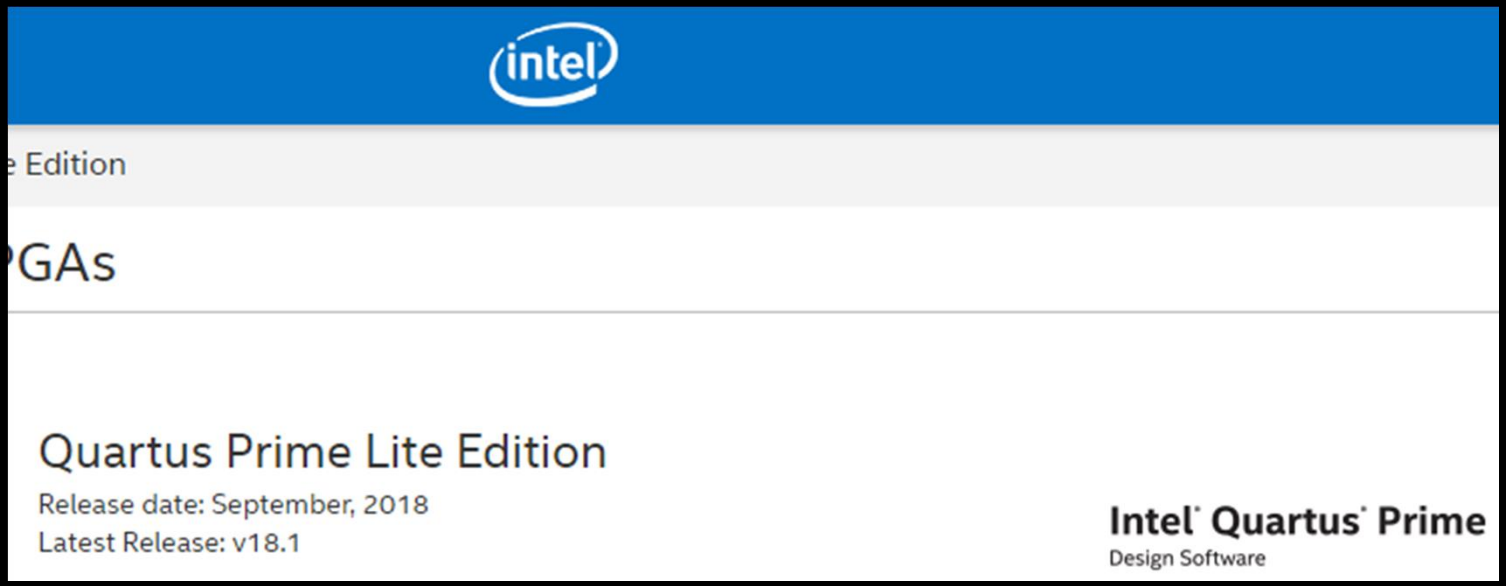
CPU: minimum of 2 cores

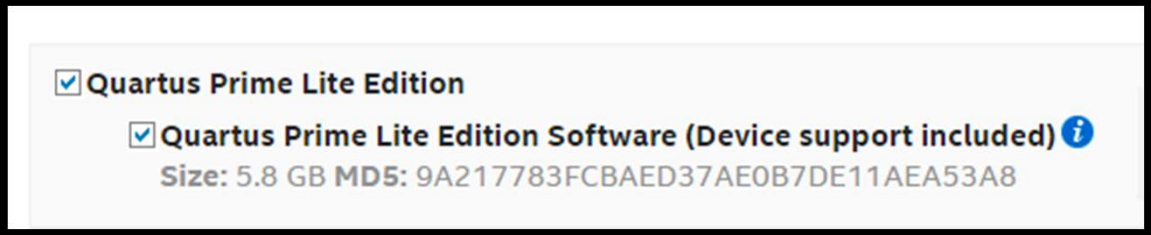
RAM: 8 GB

Disk Space: 85 GB

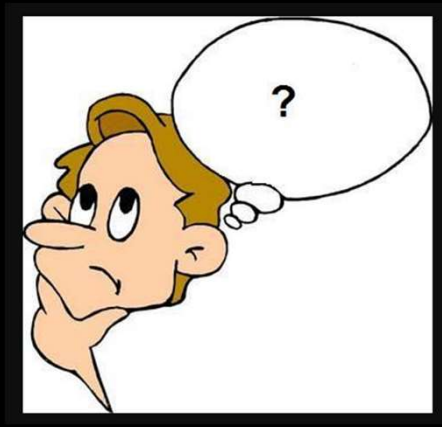
Fluxo de Projeto para Desenvolvimento

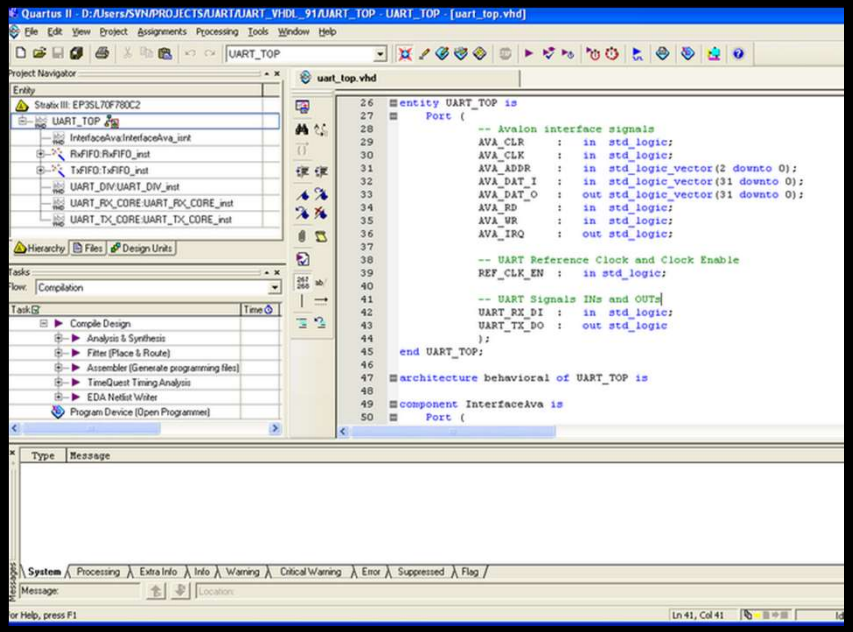
Software , Fabricantes

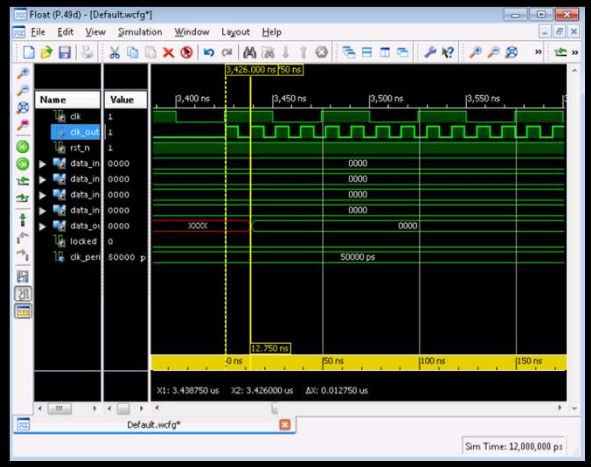


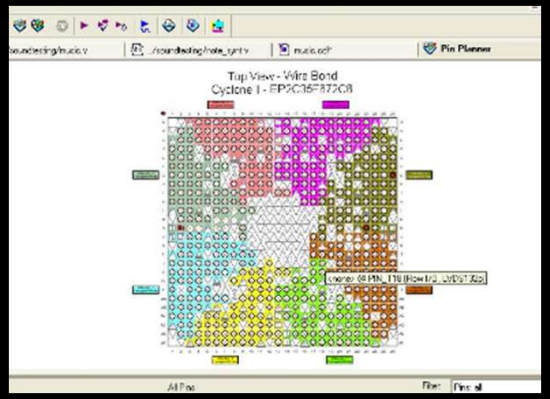


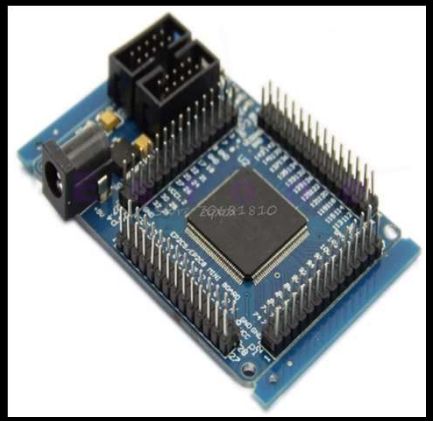
Fluxo de Projeto para Desenvolvimento



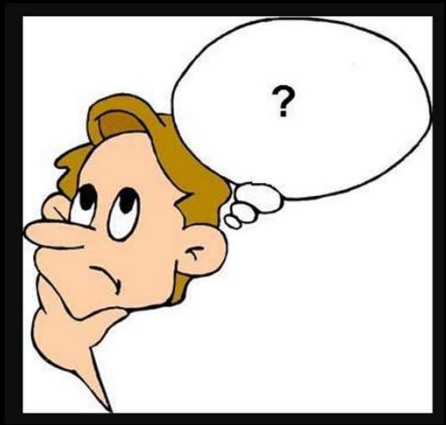








Fluxo de Projeto para Desenvolvimento – Didático



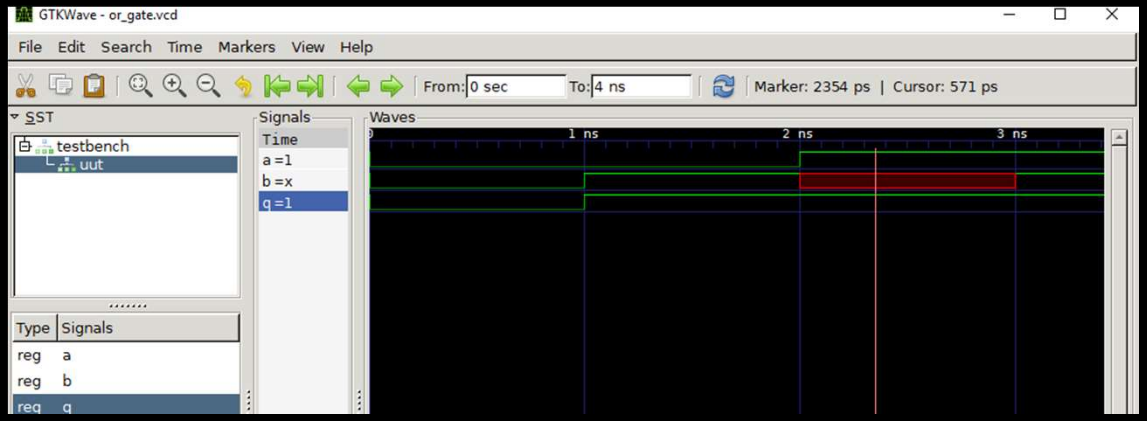
D:\download\ghdl\ghdl_036_mingw32\bin\or_gate.vhd - Notepad++

Arquivo Editar Localizar Visualizar Formatar Linguagem Configurações Tools

or_gate.vhd | testbench.vhd | half_adder.vhd | half_adder_simple_tb.vhd

```
1  -- Simple OR gate design
2  library IEEE;
3  use IEEE.std_logic_1164.all;
4
5  entity or_gate is
6  port(
7    a: in std_logic;
8    b: in std_logic;
9    q: out std_logic);
10 end or_gate;
11
12 architecture rtl of or_gate is
13 begin
14   process(a, b) is
15   begin
16     q <= a or b;
17   end process;
18 end rtl;
19
```

```
4\bin>ghdl -r half_adder_simple_tb --vcd=or_gate.vcd
```



Fluxo de Projeto para Desenvolvimento – Didático

GHDL

GHDL is an open-source simulator for the VHDL language. GHDL allows you to compile and execute your VHDL code directly in your PC.

GHDL runs on Linux, Windows and Apple OS X.

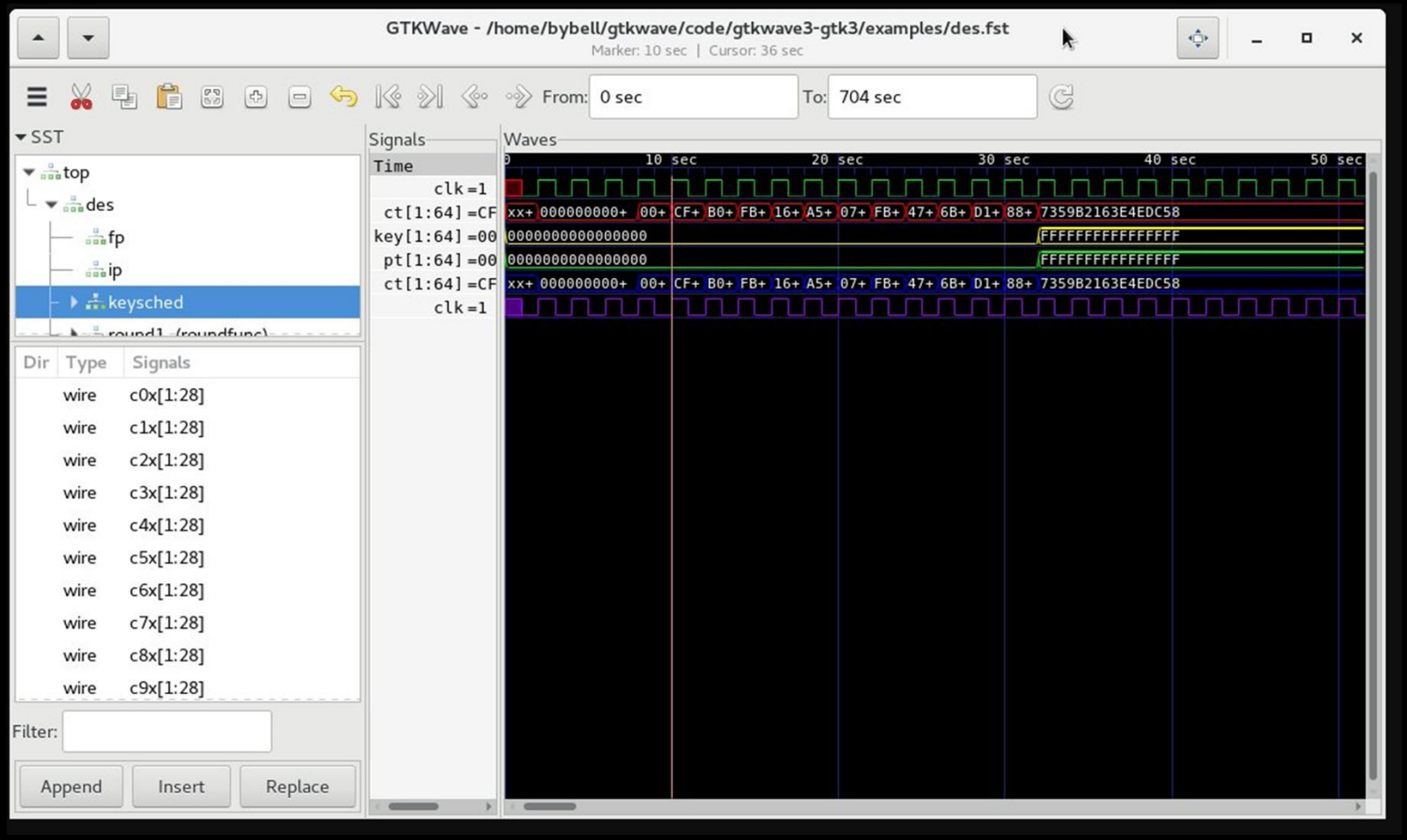
D:\download\ghdl\ghdl_036_mingw32\bin\or_gate.vhd - Notepad++

Arquivo Editar Localizar Visualizar Formatar Linguagem Configurações Tools

or_gate.vhd | testbench.vhd | half_adder.vhd | half_adder_simple_tb.vhd

```vhdl
-- Simple OR gate design
library IEEE;
use IEEE.std_logic_1164.all;

entity or_gate is
port(
  a: in std_logic;
  b: in std_logic;
  q: out std_logic);
end or_gate;

architecture rtl of or_gate is
begin
  process(a, b) is
  begin
    q <= a or b;
  end process;
end rtl;
```

Fluxo de Projeto para Desenvolvimento – Didático

GTKWave is an analysis tool used to perform debugging on Verilog or VHDL

Unix and Win32